

A modular functional–structural plant model integrating canopy architecture, carbon allocation, water transport and fruit growth for perennial fruit crops

J. Zhu^{1,a}, J. Bristow², O.-A. Chuang², X. Yang³, A. Rampadarath², and F. Rojo⁴

The New Zealand Institute for Plant and Food Research Limited (Plant and Food Research), New Zealand:

¹Marlborough, ²Auckland, ³Lincoln, ⁴Havelock North

Abstract

Functional–structural plant models, that represent plant architecture and organ function have potential to guide interventions that improve fruit productivity and quality and enhance resource and energy use efficiency. However, most existing models are species-specific, lack modularity, or cannot fully capture the dynamic interactions among plant architecture, physiology, and fruit development across scales. Here we present FruitCropXL, a generic modular functional–structural model for fruit crops, developed within GroIMP platform using the eXtended L-system (XL). FruitCropXL conceptualizes plants as collections of objects including buds, flowers, fruits, leaves, internodes, fine roots, and structural roots, complemented by classes including Phenology, Environment, and Resource Arbitrator. Originating from GrapevineXL, FruitCropXL simulates key physiological processes at the individual leaf level, including light interception, photosynthesis, and transpiration. It further models water uptake from different soil layers, water transport from soil to leaf based on water potential gradients and phloem sugar concentration by tracking the carbon dynamics of each organ. As a key advancement, FruitCropXL integrates a biophysical virtual fruit module, enabling mechanistic simulation of diverse fruit types and their growth processes, including sugar and acid metabolism, dynamically linked to whole-plant carbon and water status. The model features a flexible modularization framework, allowing users to toggle between different species (e.g., grapevine and apple), module versions (e.g., root complexity), and functionalities such as light interception and carbon transport. This enables users to customize simulations from a single organ to the whole plant, and from hourly to seasonal timescales, within a single configuration file. FruitCropXL is equipped to handle both static and dynamic canopy architecture scenarios starting at any point in a growing season. It can also accommodate different canopy and root architectures, which can be integrated either through csv files from our newly developed shoot and root architecture generators, or by selecting predefined architecture types within the model.

Keywords: Functional-structural plant model, perennial fruit tree, virtual fruit model, precision horticulture, carbon and water transport

INTRODUCTION

Modern orchard production faces increasing pressure to improve efficiency, sustainability, and product quality in response to challenges such as climate change, resource limitations, labour shortages and evolving consumer demands (FAO, 2019). Modelling is an essential tool for designing efficient orchard systems (Marvin et al., 2022), as it can help us to improve our understanding of plant physiology. Specifically, functional–structural plant models (FSPMs) illustrate how environmental conditions, tree architecture,

^a E-mail: junqi.zhu@plantandfood.co.nz

and management strategies influence growth and fruit production. By enabling researchers to test hypotheses and explore "what-if" scenarios in silico, FSPMs can support the design of field experiments and inform the refinement of management practices thus advancing precision horticulture that maximizes productivity while minimizing environmental impacts.

The emergence of artificial intelligence, computer vision, and robotics is creating new opportunities for horticulture, including the development of digital twins that can provide dynamic, real-time representations of orchard systems. However, a key knowledge gap remains there is a lack of generic, modular plant models that can accurately simulate the complex interactions between plant architecture, physiology, and fruit development across different species and management scenarios, and that can be effectively integrated into digital twin frameworks.

Here, we present FruitCropXL as a step toward closing this gap. Our work demonstrates a flexible, multi-scale functional-structural model capable of simulating organ- to whole-plant-level processes in fruit crops, supporting scenario analysis and virtual experimentation for research and digital horticultural system development. This advancement provides a critical modelling component for future digital twins in horticulture, with the potential to accelerate both scientific discovery and practical innovation.

MODEL DESCRIPTION

FruitCropXL, a generic functional-structural model tailored for Fruit Crops (Figure 1) was developed in the GroIMP modelling platform (<https://grogra.de/>) using the eXtended L-system and relational growth grammar language (Hemmerling et al., 2008; Kniemeyer, 2008). FruitCropXL conceptualizes plants as collections of objects including buds, flowers, fruits, leaves, internodes, fine roots, and structural roots, complemented by classes such as Phenology, Environment, and Resource Arbitrator (Zhu et al., 2018a; Zhu et al., 2019). Users can flexibly configure which objects are included in the initial conditions. The model then executes in three main steps: First, it loads the configuration file and initializes all modules, objects, and parameters using factory methods. Second, it updates the state variables of all organs and objects to reflect environmental and physiological changes. Finally, if dynamic growth is enabled, the model develops new organs according to the specified growth parameters. The model also enables users to define custom update and development rules, using either the existing functions or by creating new methods. The physiological processes in the model are divided into five main components:

(a) Radiation Interception

The GroIMP platform's radiation model computes light distribution and absorption based on the optical properties of plant organs (Hemmerling et al., 2008). This model considers the number, location, and intensity of light sources, and the direction, absorption, reflection, and transmission of light rays across plant organs. Radiation interception is computed in daytime steps, capturing the energy absorbed by every plant organ in the scene. Users can select from various light types, including spectral, directional, spotlight, and customizable physical light sources. In this model, an array of 72 directional lights represents diffuse radiation, and a moving directional light source simulates direct sunlight, with its position adjusted for day length, azimuth, and solar angle based on Reda and Andreas's (2004) solar positioning algorithm, using the Java solar position package (Brunner, 2024). Clear-sky direct, diffuse and near-infrared intensities are calculated using the REST-2 model (Gueymard, 2008), with diffuse radiation fractions refined by the hourly global-to-theoretical clear-sky radiation ratio (Erbs, 1982). The platform provides both provides both central processing unit (CPU) and graphics processing unit (GPU) based light models (Henke and Buck-Sorlin, 2017). The GPU-based light model supports simulations

across the full light spectrum, whereas the CPU-based light model is limited to simulating three light channels.

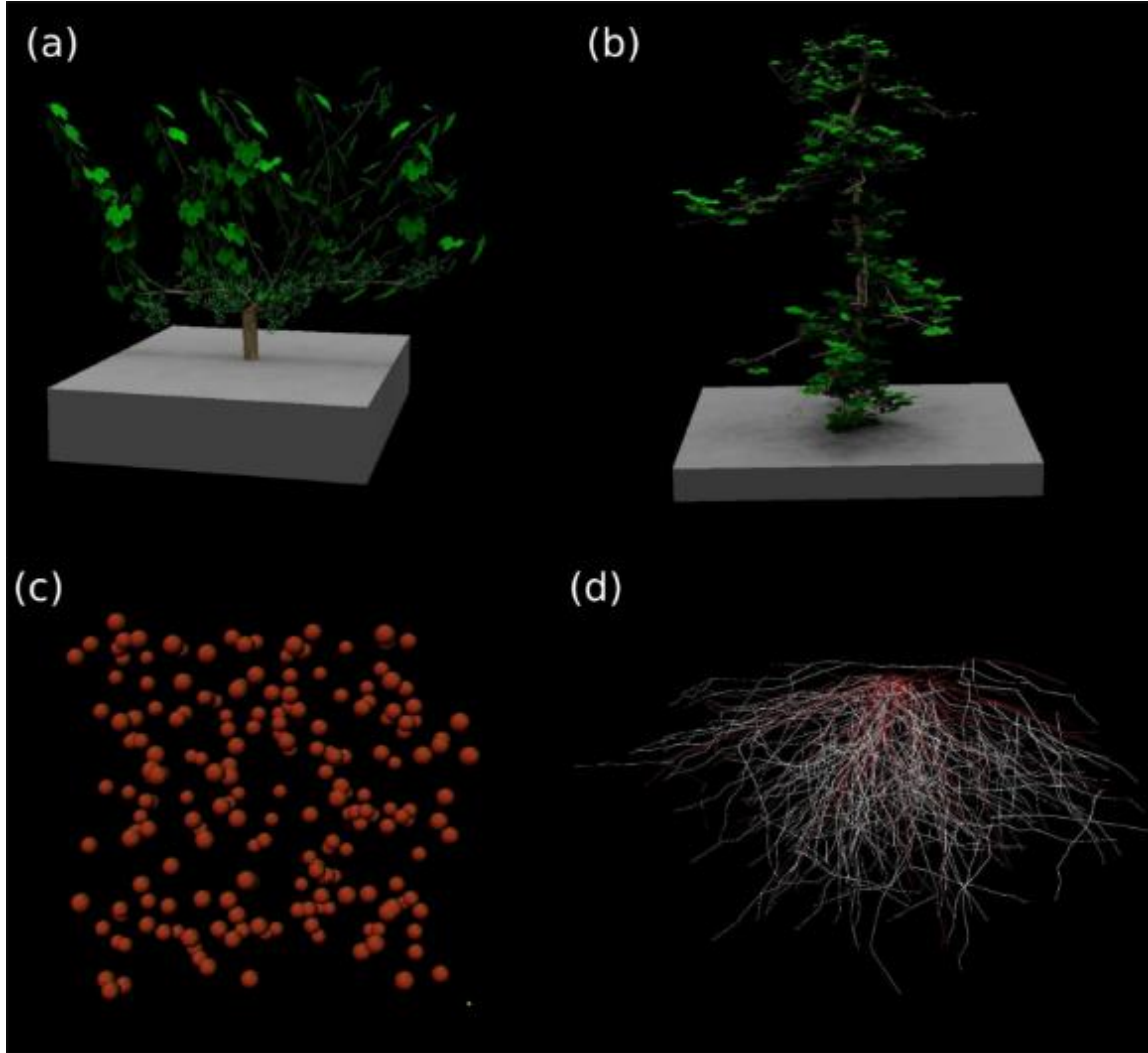


Figure 1. Illustration of four different model configurations: (a) a simulation of grapevine berry growth within a two-cane pruning system; (b) a tall spindle apple tree for studying the effects of tree architecture on light distribution, (c) a simulation of a fruit population comprising 200 apples with diverse initial traits to examine population variability in fruit characteristics, and (d) a three dimensional root architecture simulation to investigate the influence of fine root distribution on water uptake.

(b) Leaf Gas Exchange and (c) Water Transport

The model integrates an extended Farquhar, von Caemmerer, and Berry (FvCB) module for calculating potential photosynthesis, transpiration, leaf temperature, and stomatal conductance under various atmospheric conditions (Yin and Struik, 2009), along with a Tardieu–Davies water transport module. This transport model simulates water movement from soil to roots, through the xylem, and along internodes to leaves, using an electrical resistance analogy and variable hydraulic conductance to calculate water potential across organs (Tardieu et al., 2015; Zhu et al., 2018b; Zhu et al., 2021). The modules are dynamically integrated: the FvCB module provides the Tardieu–Davies module with potential stomatal conductance and transpiration. In turn, the Tardieu–Davies module supplies the FvCB module with adjusted stomatal conductance and transpiration rates, calculated via a numerical solver balancing water flux with adjusted leaf transpiration. Users

can choose stomatal conductance adjustments based solely on leaf water potential or additionally on xylem abscisic acid concentration. This feedback allows for recalculation of leaf temperature and photosynthesis based on actual transpiration rates (Zhu et al., 2018a).

(d) Carbon Allocation

The model includes two carbon allocation approaches: a common assimilate pool model and a phloem carbohydrate transport model (Zhu et al., 2021). In the assimilate pool model, carbohydrates from leaves, stems, and roots are loaded into the phloem and translocated to sink organs based on unloading capacities, assuming an equilibrium of loading and unloading rates for each hour (Baldazzi et al., 2013). The phloem/xylem coupled carbon transport model, implemented in FruitCropXL, combines analytical and computational techniques, simulating plant structure at the metamer level with internodes as conduits and lateral organs as sources and sinks. This approach allows detailed carbon transport simulations, with transport equations solved analytically and iteratively refined to account for concentration-dependent sink and source activity (Seleznyova and Hanan, 2018).

(e) Organ Development

The model uses the phytomer comprising an internode, leaf, flower or fruit, and lateral buds as the basic shoot architecture unit. Developmental processes vary by species, and are customized to their growth patterns. In the apple module of FruitCropXL, buds initiate new shoot apices based on the combination of bud function (vegetative or fruiting) and shoot type (long shoot or spur). Fruiting buds produce sympodial bourse shoots with rosette leaves and axillary shoots, while vegetative buds give rise to monopodial shoots that grow continuously from a single, persistent apex. Shoot length is determined by both shoot type and the vertical position of the bud within the canopy, as observed in our data. Flowers and young fruitlets (<30 days after flowering) are represented by the flower class to account for their carbohydrate demand. After 30 days, they are reclassified as fruit organs, at which point the virtual fruit model is activated to simulate hourly sugar and water uptake (Fishman and Génard, 1998; Génard et al., 2007).

Modularization Framework

Over time, we have enhanced FruitCropXL with a modularization framework that enables the toggling between different species (e.g., grapevine and apple), module versions (e.g., simple or complex root modules), and various functionalities such as light interception and carbon transport. This modular framework provides users with the flexibility to customize simulations, ranging from modeling a single organ to simulating an entire plant, and from one hour to an entire season, all within a single configuration file. The model is capable of handling both static and dynamic canopy architecture scenarios starting at any point within a growing season. It can also accommodate different canopy and root architectures, which can be integrated through CSV files generated by our newly developed shoot (Rojo et al., 2024) and root architecture generators (Bristow et al., current proceeding), or alternatively through direct coding of architecture types within the model itself.

The initial configuration setups are designed to be highly flexible, allowing users to define parameters such as planting densities, row orientation, climatic conditions, canopy architecture and initial root biomass etc., thereby ensuring realistic initial conditions tailored to specific species or agronomic requirements. Configuration files permit precise adjustment of plant and optional soil parameters for each simulation, thereby facilitating control over model complexity. For instance, fruit models can shift from simple to complex representations that utilize detailed virtual fruit packages. Similarly, the soil model can be adjusted to include either a basic layered representation or a detailed three-dimensional grid. The complexity of root and leaf architectures can also be customized, from simplified conceptual layers to intricate 3D structures that reflect natural variability.

Users have the option to conduct specialized simulations focused on specific components, such as isolating a single leaf, root system, or a subset of fruits without the need to simulate the entire plant structure. Furthermore, users can design custom plant architectures or employ species-specific modules to improve simulation accuracy for particular crops.

Environmental conditions such as temperature, humidity, light intensity, and soil water conditions can be incorporated from external data files to simulate the influence of dynamic environmental factors on organ growth. Various output options and visualizations are available for individual organs, plant and field level summaries. Additionally, Doxygen is used to generate comprehensive, web-based documentation that includes detailed descriptions of the model structure, classes, and methods (Van Heesch, 2008).

RESULTS

In this study, we demonstrate the functionality of the model using two case studies: (1) a two-cane pruned grapevine with a leaf area of 7.5 m², grown in rows spaced 2.4 meters apart and 1.8 meters between plants (Figure 2); and (2) a planar cordon apple tree with a leaf area of 25 m², grown with 2-meter row spacing and 3-meter plant distance (Figure 3).

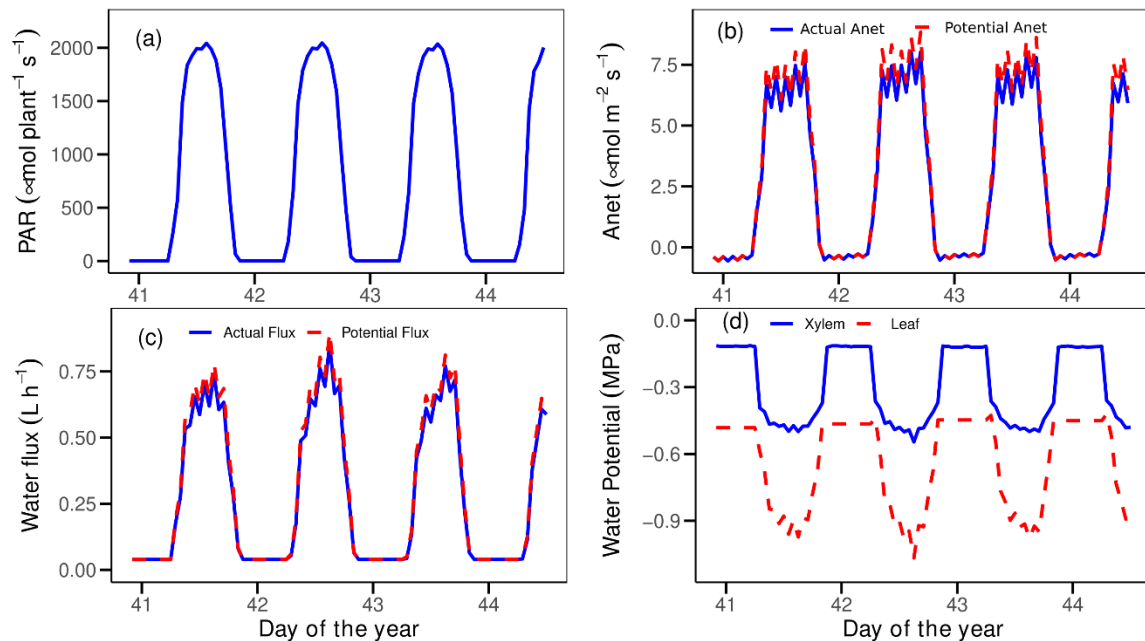


Figure 2. Absorbed photosynthetic radiation per plant (a), net photosynthesis per unit of leaf area (b), water flux per plant (c), xylem and leaf water potential dynamics (d) of two-cane pruned grapevine with a leaf area of 7.5 m² grown in rows spaced 2.4 meters apart and 1.8 meters between plants. Blue lines in panel b and c represent the actual net photosynthesis and water flux, respectively, while red lines represent the potential values without accounting for the water stress.

In the grapevine case study, the climate data from Marlborough (2016) New Zealand was used, and soil water potential was set to -0.12 MPa based on soil water content records at the meteorology station. Simulations were conducted to illustrate the model's ability to simulate light interception, photosynthesis, transpiration, and water potential gradients. The two-cane grapevine system intercepts approximately 25% of incoming radiation, with interception peaking at about 2000 μmol per plant per second at noon (Figure 2a). On a sunny day, the average photosynthesis rate reaches 7.5 μmol CO₂ per square meter of leaf area per second, producing approximately 62 grams of carbohydrate daily (Figure 2b). Concurrently, water flux peaks at around 0.75 liters per hour (Figure 2c). Xylem water potential decreases from -0.12 MPa at dawn to -0.5 MPa by noon, whereas leaf water

potential, initially low at -0.45 MPa during the night due to nighttime transpiration, drops further to -1 MPa at noon (Figure d). Under the current soil water conditions, differences between actual and potential photosynthesis remain negligible.

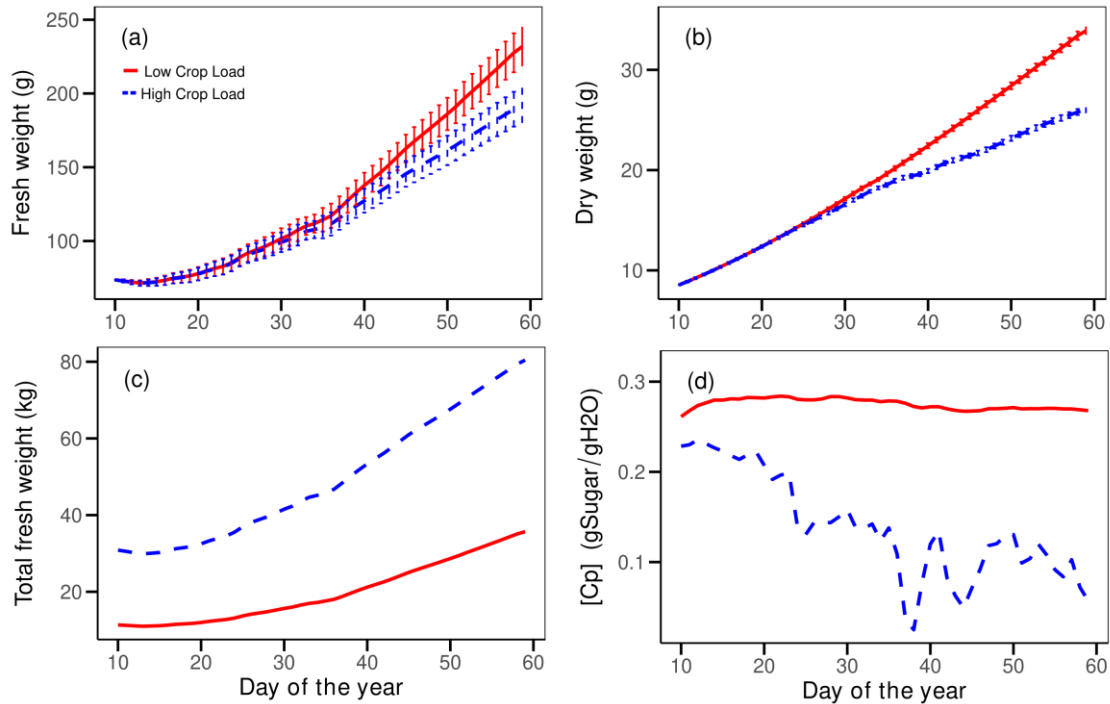


Figure 3. Effects of crop load on fruit fresh weight, dry weight, total fruit fresh weight, and phloem sugar concentration Cp. Simulations were conducted under two two crop loads: 420 fruits (blue lines) and 150 fruits (red lines) with a leaf area of 25 m^2 per tree. The simulations were based on a planar cordon apple tree grown with 2-meter row spacing and 3-meter plant distance. Error bars represent the standard deviation among individual fruits.

In the apple case study, the climate data from Havelock North (2022) New Zealand was used, and soil water potential was set to -0.05 MPa based on soil water content records at the experiment site. Simulations were conducted to examine the effects of contrasting crop loads on fruit development and carbohydrate allocation over the fruit ripening period from the 10th January to 1st March (Figure 3). Under both low (150 fruits per tree) and high (420 fruits per tree) crop load scenarios, fruit fresh and dry weights increased steadily throughout the season. However, individual fruit fresh weight was consistently higher under low crop load, reflecting reduced competition for assimilates (Figure 3a). A similar trend was observed for dry weight accumulation, with low crop load trees producing heavier individual fruits compared to those under high crop load (Figure 3b). Despite these differences in individual fruit size, the total fresh weight of fruit per tree was substantially greater under high crop load, due to the larger number of fruits (Figure 3c). Phloem sugar concentration ([Cp]), which serves as an indicator of carbohydrate supply to developing fruits, remained high and stable in trees with low crop load but declined markedly in the high crop load treatment, particularly from day 20 onward. This decline suggests that increased sink demand under high crop load results in limited carbohydrate availability per fruit (Figure 3d). Collectively, these simulations demonstrate how the model can be applied to assess the physiological trade-offs associated with crop load management in apple.

CONCLUSION

We have developed FruitCropXL, a flexible and advanced functional-structural fruit crop model designed for various fruit crops, built within the GroIMP platform using

eXtended L-system (XL). FruitCropXL represents 3D plant architecture and simulates physiological processes including light interception, photosynthesis, transpiration, water transport, and fruit growth. Enhanced with a modular framework, FruitCropXL allows customization for different crop species, and functionality toggles, enabling users to simulate conditions from individual organs to entire plants across various timeframes. The case studies demonstrate the model's effectiveness in simulating resource allocation and physiological responses under varying management and environmental conditions. These capabilities make FruitCropXL a valuable tool for supporting precision crop management decisions and optimizing horticultural production systems.

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